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REVIEW

A mixed-method review on greenwashing: its manifestation forms and "3p" greenwashing's theoretical framework

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Abstract

The deceptive practice of greenwashing (GW) has successfully misled stakeholders, creating a false perception of an organization's commitment to sustainability. Greenwashing behavior (GWB) is a crucial aspect that must be clarified before promoting sustainable development. Hence, how to scientifically understand GWB has become the core issue for enterprises to carry out green practices under the goal of environmental sustainability, especially in terms of its manifestations. Based on this, this study aims to use 553 peer-reviewed articles published from 2001 to 2024 as data to systematically analyze existing research on GWB using quantitative and qualitative methods, and construct a comprehensive GWB theoretical framework. Firstly, in the process of quantitative statistical literature review, four main research dimensions are formed to address the issue of fuzzy classification in previous studies. Secondly, a qualitative review will be conducted on the main content of the four research dimensions to provide GWB with a more structured understanding. Meanwhile, seven core elements of GWB manifestation were revealed. Finally, the above content is summarized and form a theoretical framework for GWB based on the perspectives of people, process, and purpose (3P). The proposed 3P theoretical framework of GWB not only advances the theoretical discourse by consolidating insights on definitions, driving factors and underpinning theories, governance strategies and classifications and consequence, but also provides practical tools for managers and researchers to identify, prevent, and avoid GWB, which render new perspectives in promoting ethical and sustainable practices across industries in the future.

Keywords Greenwashing behavior · Manifestation forms · Theoretical framework · Environmental research · Sustainability

Introduction

In the twenty-first century, businesses are increasingly embracing green practices to showcase their corporate social responsibility (CSR). While businesses make valuable contributions to social welfare, people complaints about its greenwashing (GW) are common (Wu et al. 2020). In recent years, GW has become a highly concerned social issue both domestically and internationally in the context of sustainable development. For example, the European Commission has stated that approximately 42% of companies in the EU market are suspected of GW (Xie et al. 2024). Greenwashing is a combination of the words "green" and "whitewashing" (Martinez et al. 2020). It is a form of empty rhetoric that masks genuine social responsibility efforts, helping businesses create a misleadingly environmentally friendly green image. For example, Coca-Cola company claims to reduce water consumption by 4% annually to conserve water resources, but it remains silent on its actual water usage (Chen and Dagestani 2023). Huang et al. (2020) mentioned that emissions cheating was a common practice among many automotive industry giants during the Volkswagen GW scandal. The frequent occurrence of GW has gradually become an obstacle to the development of an ecological civilization.

GW is a phenomenal and strategic concept, while greenwashing behavior (GWB) is its concrete expression of action levels. GWB describes as a deceptive marketing or communication practice with various manifestation forms. GWB can have negative consequences, such as consumer

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skepticism toward green claims (Zhang et al. 2022; Farooq and Wicaksono 2021), a decline in brand loyalty (Rahman and Bang Nguyen-Viet 2022), and the erosion of government-corporate relationships (Li et al. 2022). Despite the implementation of stringent environmental policies and regulations by countries (Guo et al. 2017), which have resulted in some initial improvements in controlling environmental issues, the persistence of GWB necessitates ongoing vigilance. This is because there is a widespread phenomenon of information asymmetry in the market (Li et al. 2023b, a), and the complexity of environmental information (such as carbon footprint, circular economy indicators, etc.) further exacerbates the consequences of information asymmetry (Tao et al. 2024; Abbasi and Scakyüz, 2024). This results in environmental problems not being detected in a timely manner, delaying the timing of problem handling, and masking their harmfulness, seriously interfering with the effectiveness of government regulation. Therefore, this study takes the social problem of greenwashing behavior as the starting point, and forms a comprehensive scientific understanding by discovering the complex causal relationships behind it, providing theoretical guidance for regulating corporate environmental behavior and achieving high-quality green development.

GWB relevant research has attracted considerable attention from scholars. Previous studies on GWB were limited to specific aspects or impacts of the practice. For example, existing qualitative studies have mostly concentrated on its definitions (Nemes et al. 2022), driving factors (Delmas and Burbano 2011; Tateishi 2018; Chen et al. 2022), and impacts (Kahraman and Kazancoglu 2019), or combinations of various aspects like antecedents and consequences (De Jong et al. 2018), concept and manifestations (Netto et al. 2020). Whereas, the quantitative research has focused on studying the impact of GWB on consumers (Wang et al. 2020a, b; Farooq and Wicaksono 2021), corporate performance (Testa et al. 2018;), and governance measures (Huang et al.,2022; Nardi 2022). These papers focused on a specific area of GWB. They had not formed a cohesive understanding regarding GWB, particularly from clarifying manifestation forms of GWB in a comprehensive manner. Unlike other existing literature reviews (Marquis et al. 2016; Yang et al. 2020; Netto et al. 2020), this study adopts an inductive and integrative approach to systematically analyze the evolution of GWB research. By doing so, it addresses a critical gap in the literature—namely, the lack of a comprehensive theoretical framework that encapsulates the multidimensional nature of GWB. Below are the research questions that guide us in this review process.

(1) What are the key dimensions of GWB research, and which industries are most significantly affected or frequently examined within this domain? Furthermore, which sectors have emerged as hotspots in recent GWB studies?

(2) What are the core elements that characterize the manifestation forms of GWB, and how are these elements conceptualized and operationalized in existing literature?

(3) Based on a systematic synthesis of the above findings, what comprehensive theoretical framework can be proposed to deepen the understanding of GWB and guide future empirical investigations?

As a result, a mixed-method review was adopted in this paper. It delved into the multifaceted concept of GWB including drivers across external, organizational, and individual dimensions and suggests governance strategies from government, corporate, societal angles, and other aspects. Meanwhile, the 3P (people, process, and purpose) framework originates from organizational behavior (Wang et al. 2020a, b), emphasizing the explanation of complex behavior from three aspects: individual or organizational characteristics, behavioral motivation, and behavioral pathways. It provided a systematic paradigm for GWB, starting from the triple path of "the person who acts, the cause of the behavior, and the way the behavior occurs". The proposed framework would serve as a background for qualitative or quantitative research on GWB issues in different industries in the future. Based on these three aspects, not only does it enrich the existing theoretical research system of environmental management and opportunistic behavior, but it also provides a basis for regulation design multidimensional governance strategies. These strategies can help reduce the incidence of GWB and enhance corporate sustainability transparency. Therefore, the study can be broadly divided into the following three aspects: (1) retrieval and statistical analysis of GWB related literature via quantitative method. (2) qualitative review the research on GWB in terms of definitions, driving factors, governance strategies and classifications and consequence, and (3) clarify the core elements of manifestation forms of GWB and formulate a comprehensive theoretical framework for GWB.

Research methods

In this study, we conducted a mixed-method approach. By combining quantitative, such as document statistics, and qualitative methods, such as content analysis, this study reveals the research trends of GWB at a macro level while deepening the theoretical understanding of the research content of GWB at a micro level. Therefore, the proposed approach is suitable for exploring the complex and multidimensional research topic of GWB (Leech and Onwuegbuzie 2007). Firstly, this paper adopted a quantitative method to search for relevant publications in the Web of Science database and conducted article statistics based on relevant criteria. Secondly, a comprehensive literature review was conducted using qualitative methods to depict the development

and implementation of GWB and focused on analyzing the manifestation forms of GWB. Finally, a comprehensive GWB theoretical framework was formed. The specific stages are as follows:

In Stage 1, document search was performed using the "keywords/title/abstract" fields of the Web of Science Core Collection. While Scopus provides broader journal coverage, the choice of Web of Science was based on its comprehensive coverage of high-impact, peer-reviewed literature in related fields, making it a reliable source for identifying influential studies on greenwashing. Therefore, Web of Science was considered more suitable for ensuring the quality and relevance of literature. To maintain focus and ensure data relevance, the publication time span was restricted to 2001–2024. The primary search terms used were "greenwashing" and "greenwashing behavior", as these directly correspond to the core concept under investigation. While broader terms such as "deceptive environmental marketing" or "corporate green claims" could provide additional context, they were not included to maintain specificity and conceptual clarity in this stage of data collection. These keyword sets were employed to capture relevant variations of the terms, initially resulting in the retrieval of 825 articles.

In Stage 2, the research direction was limited to fields such as "management science," "social science," "environmental science," and "economics". This is because greenwashing is a multifaceted socio-organizational phenomenon that spans strategic business behavior, public perception, environmental impact, and regulatory-economic systems. Based on this, 738 articles were retrieved. In addition, to ensure the relevance of the topic, a manual filtering method was adopted to briefly review the abstracts and content of the obtained articles, eliminating or adding irrelevant publications. Specifically, this involves reviewing the title and abstract of the article, and if necessary, examining the main content of the article to determine their direct connection to greenwashing. Publications that only mention the term 'greenwashing' without contributing to GWB are excluded. For instance, the study by Luckmika et al. (2019) was excluded as their research content is unrelated to GWB despite mentioning the term in their paper. In total, 348 publications are identified for the discussion of the research content in this paper.

In Stage 3, In order to conduct a comprehensive review of the selected articles, this study is divided into three analytical stages. Firstly, a descriptive statistical analysis was performed on the 553 selected articles, covering publication trends over time, journal distribution, and industry focus. In addition, VOSviewer was used to form topic word clustering. The data were systematically recorded to ensure accuracy and consistency in classification, thereby enhancing the reliability of the quantitative results. Second, a qualitative content analysis was conducted to systematically identify and interpret recurring themes related to GWB. Specifically, we adopted the content analysis approach to review the article abstract and, if necessary, the full text to extracted key concepts and classified them into higher-order themes. This process enables us to identify core research dimensions, such as antecedents, impacts, and governance of GWB, and research hotspots. Finally, through a comprehensive review of literature from different dimensions, a theoretical framework was developed to understand GWB.

Based on the above three stages, Fig. 1 illustrates the general research approach in this paper.

Overview of selected GWB publications

Quantitative analysis was conducted on the publications selected in the first stage and the second stage, examining the basic research content of each article, including journal distribution and articles distribution by research subjects.

Distribution of articles by journal

Due to the large number of journals, we list the journals that published two or more articles, as shown in Fig. 2. The top five journals in terms of publication count are "Sustainability", "Business Strategy and the Environment", "Journal of Cleaner Production", "Journal of Business Ethics," and "Corporate Social Responsibility and Environmental Management," with 42, 40, 37, 32, and 29 articles, respectively. The 37 journals represented in the figure account for 58.37% of the total publication count. There are also authoritative journals such as "Academy of Management Review" (Vedantam et al. 2021) and "Production and Operations Management" (Nardi 2022) among the journals that have published one article in this field. Therefore, research on GW and related issues has also been favored by some authoritative journals.

The bibliometric analysis also reveals that the trend of the number of publications over time, which could reflect the potential of research in a particular field. Figure 3 displays the annual publication count of GWB research from 2001 to 2024. It can be observed that GWB started to receive attention in 2014, and the number of publications gradually increased. The increasing number of publications in recent years suggests that GWB is a promising topic in the field of environmental research in the future. It is expected that the publication count on this topic will continue to grow.

Industry articles distribution

Through analysis, GWB research mainly covers nine industries, including the advertising industry, food industry, energy industry, financial industry, service industry,

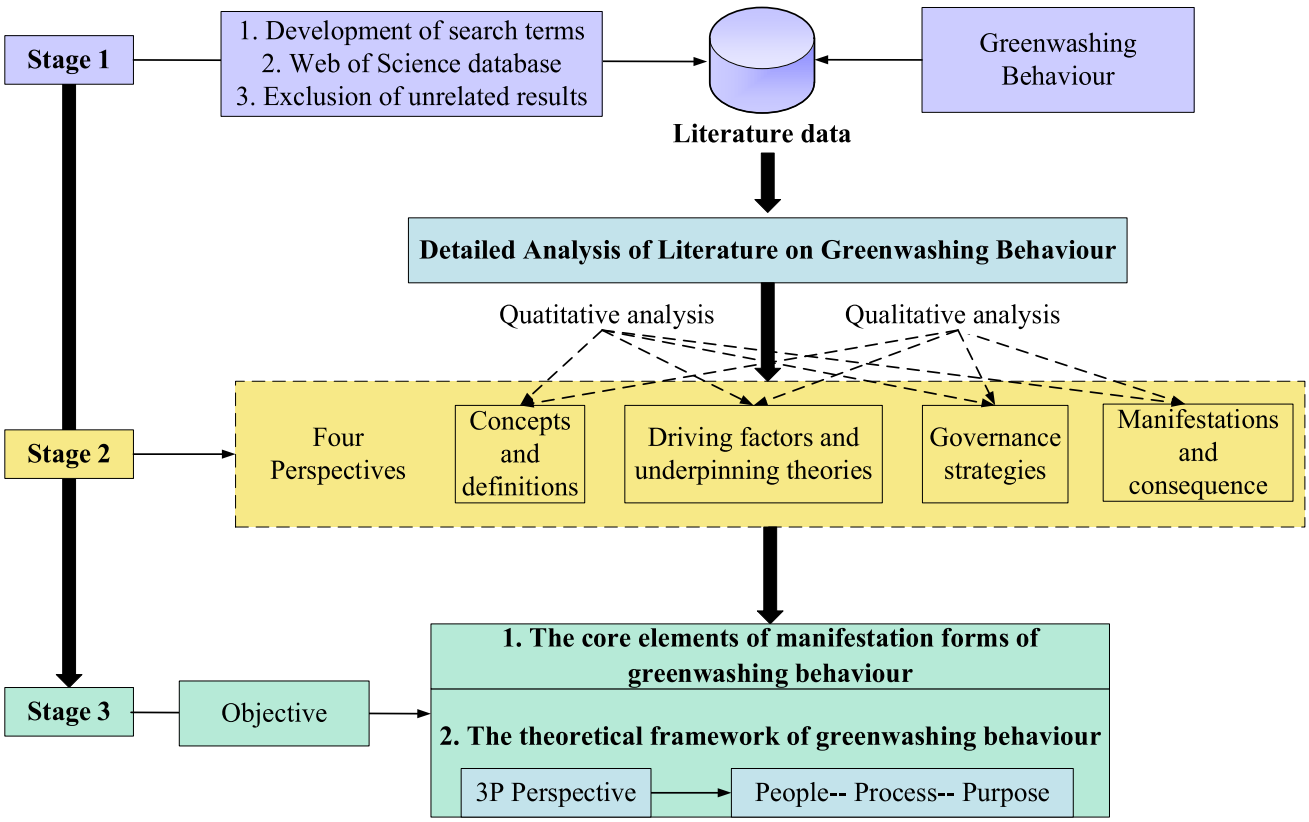


Fig. 1 Technology roadmap

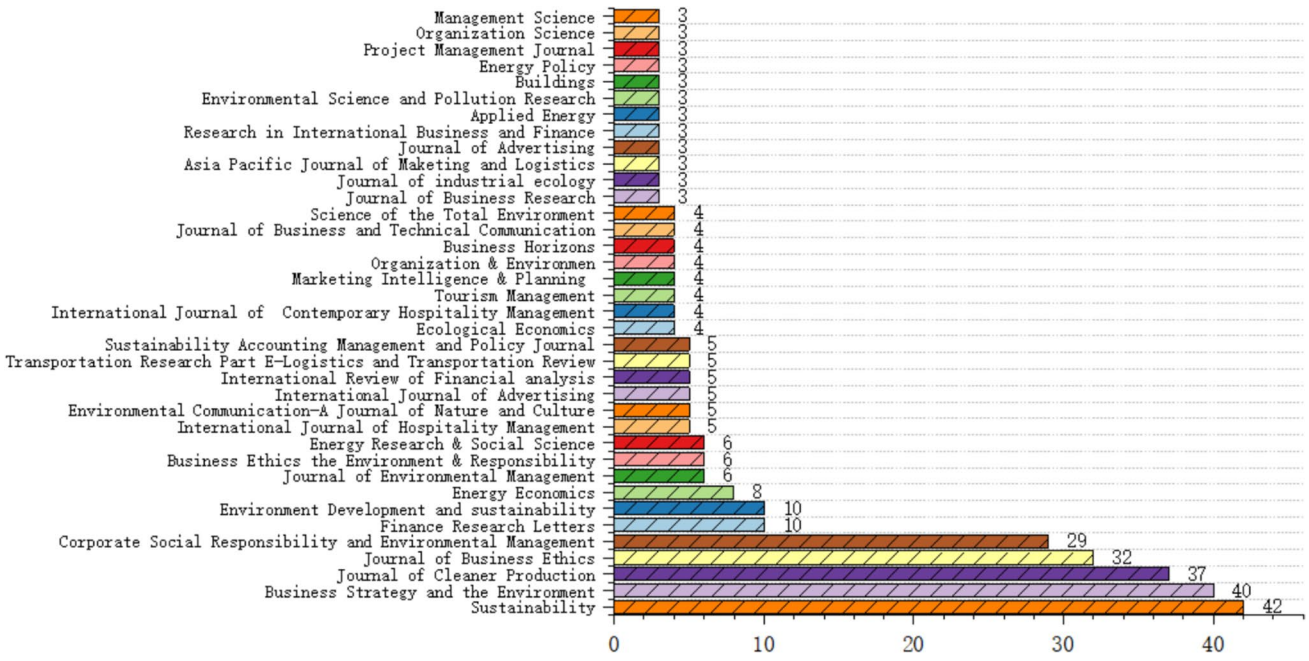


Fig. 2 Status of GWB's articles and journals

construction industry, manufacturing industry, forestry industry, and fashion industry. It is worth noting that some empirical studies use data that encompasses multiple companies without distinguishing specific industries in detail

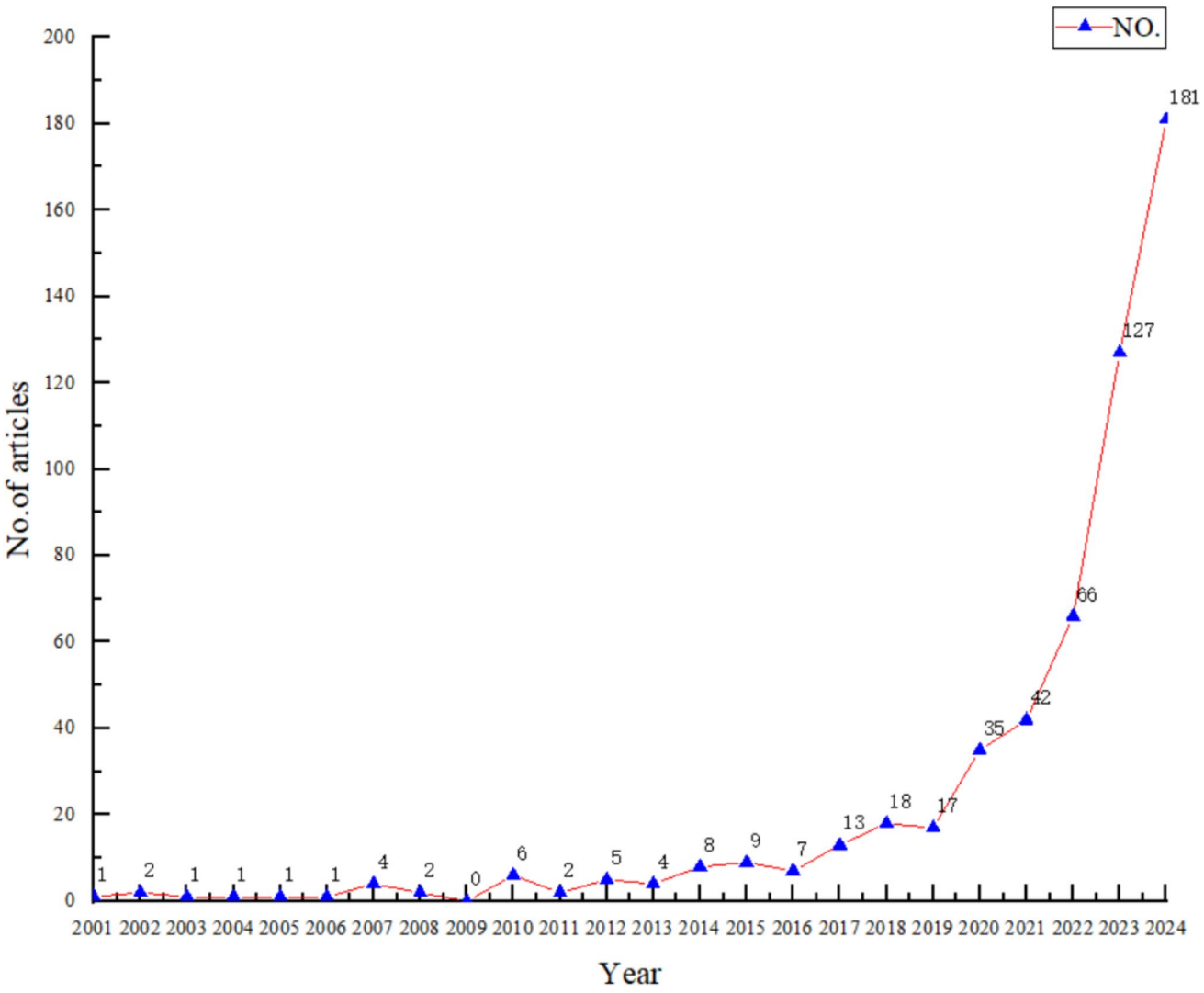


Fig.3 Distribution of GWB's articles by year

(Mateo-Marquez et al. 2022), so these articles are classified as mixed industry. Additionally, Tang et al (2020) took a different approach by focusing on GWB research at the government level, so this article is classified under the government category.

Table 1 lists the distribution of articles based on industry modules, showing that GWB issues have gradually permeated various industries. From Table 1, it could be seen that the financial industry, manufacturing industry, service industry are environmentally sensitive areas that were prone to causing GWB. It indicates that great research efforts for GWB had been made in these industries. It is worth noting that compared to the period from 2001 to 2015, the research growth rate in the construction industry has been the fastest since 2016.

Consequently, Table 2 shows some important findings from the top three industries in terms of research quantity, such as driving factors, means, and governance strategies of these three industries.

Articles allocation

From Fig. 3, it can be seen that there is an expanding trend in research on GWB. In order to better organize fragmented literature in an orderly manner, this study uses VOSviewer to identify research topic words in this field, and scientifically divides the research topics of this field based on the topic words (Sina and Hassan 2023). This provides a basis for discovering research hotspots and subsequent content analysis. The theme word clustering view of GWB is shown in Fig. 4. This graph can clearly present the three major research modules in this field, namely governance strategy research centered on "enterprise" and "greenwashing behavior", driving factor research centered on "disclosure" and

Table 1 The distribution of publications by sectors

Research sector	Period(Year)						
	2001–2005	2006–2010	2011–2015	2016–2020	2021–2023	Total	Percentage (%)
Advertising	0	1	2	6	15	24	4.34
Agriculture	1	0	1	2	16	20	3.62
Energy	1	0	3	19	35	58	10.49
Finance	0	1	2	15	34	52	9.40
Service	0	0	4	9	15	28	5.06
Construction	0	1	0	4	13	18	3.25
Manufacturing	0	2	1	19	29	51	9.22
Forestry	0	0	0	1	6	7	1.27
Fashion	0	0	1	5	13	19	3.44
State	0	0	0	1	6	7	1.27
Mixed	4	8	14	44	199	269	48.64
Total	6	13	28	125	381	553	100

Table 2 Relevant information on GWB in the three industries

Finance industry	
I. Driving factors	1. Poor quality of information disclosure (Amores et al., 2022) 2. Lack of strong supervision (Xu et al. 2022)
II. Means	1. Green bonds (Jones et al. 2020) 2. Financial products are nominally green (Popescu et al. 2021) 3. "Green" investment and financing business of enterprises (Wu and Shen 2013)
III. Governance strategies	1. Block-chain technology (Christodoulou et al. 2023) 2. Third-party certification (Zhu et al. 2023)
Manufacturing industry	
I. Driving factors	1. Relaxed regulatory environment (Zhang et al. 2022) and competition among enterprises (Arouri et al. 2021) 2. Financial pressure on enterprises (Mateo-Marquez et al. 2022) 3. Personal traits (Cao et al. 2022)
II. Means	1. False green advertising (Farooq and Wicaksono 2021) 2. Unreliable certification information (Heras-Saizarbitoria et al. 2020)
III. Governance strategies	1. The necessity of government regulation (Mateo-Marquez et al. 2022) 2. Improve the transparency of corporate information (Reck et al. 2022)
Service industry	
I. Driving factors	1. Cost savings and improve image (Gupta et al. 2019)
II. Means	1. Pseudo "green" marketing methods, e.g. towel reuse plan (Chen et al. 2019)
III. Governance strategies	1. Personal traits (Gupta et al. 2019) 2. Social supervision (Testa et al. 2018)

"pressure", and consequence and manifestation research centered on "consumer" and "claim".

This study conducted a literature review based on the themes obtained above and found that the definition of GWB has also received attention from many scholars (Nemes et al. 2022). This is because in order to systematically understand GWB, there must first be a clear conceptual definition. Previous literature has proposed different definitions of GWB, reflecting its complexity and context-dependent nature, which form the theoretical basis for subsequent research. Therefore, the content analysis indicates that GWB-related research can be divided into four major research modules: definition, driving factors, governance strategies and manifestation form and consequence. It should be noted that some articles address more than one module. For instance, Kahraman and Kazancoglu (2019) conducted a detailed examination of GWB definition and its determining factors from the perspective of consumers, thus managerial insights for businesses in formulating green marketing strategies were provided.

Table 3 present the distribution of articles according to research modules. This indicate that significant research efforts have been devoted to exploring the driving factors of GWB. It is worth noting that since 2016, there has been a

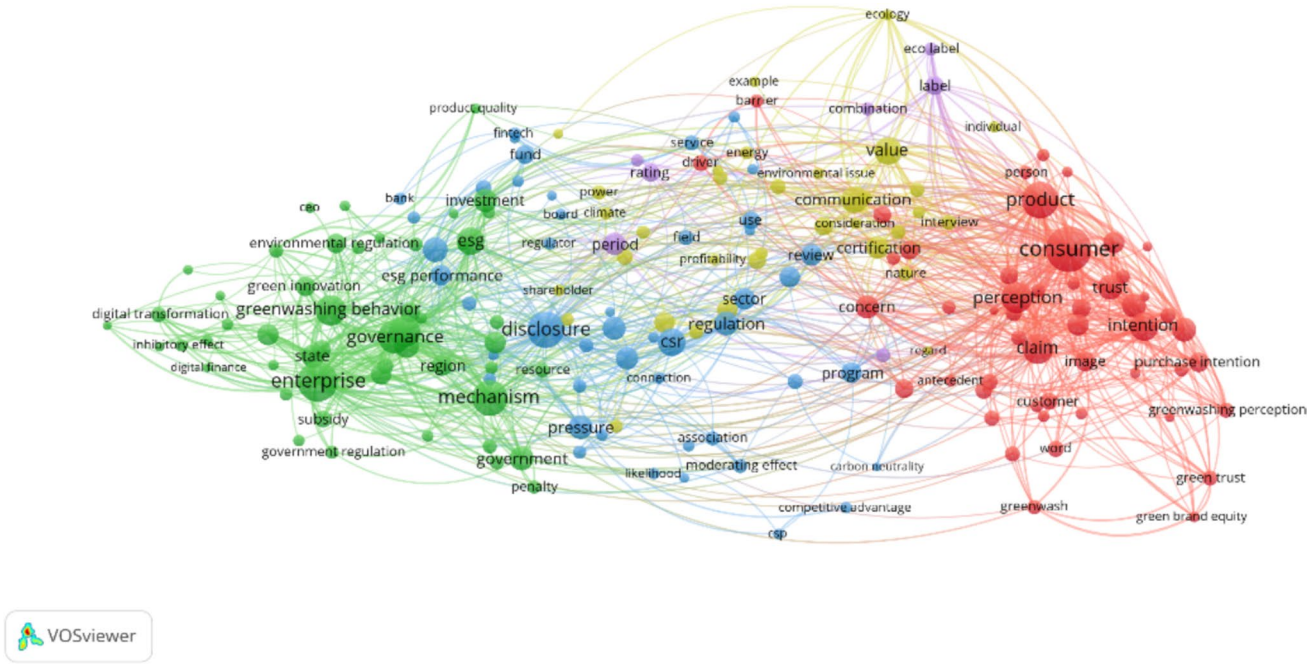


Fig. 4 Topic clustering view of GWB

Table 3 The distribution of publications by research modules

Perspective	Period(Year)						
	2001–2005	2006–2010	2011–2015	2016–2020	2021–2023	Total	Percentage (%)
Definition	1	1	1	2	11	16	2.89
manifestations and consequence	0	3	5	19	76	103	18.63
Driving factors	2	5	11	20	85	123	22.24
Governance	0	1	5	18	110	134	24.23
Mixed	3	3	6	31	134	177	32.01
Total	6	13	28	90	416	553	100

rapid increase in research on the GWB's manifestation forms and driving factors.

Literature review of different dimensions

In this section, a systematic analysis of the survey results was conducted based on the four dimensions of GWB. Thus providing a comprehensive scientific understanding for GWB existing research.

Concepts and definitions of GWB

In 1986, biologist Jay Westervelt first introduced the term GWB into real-life contexts (Netto et al. 2020). In the New Millennium English Dictionary, Wester defined GWB as "the practice of promoting environmentally friendly programs to divert attention from the unfriendly activities of organizations towards the environment." In 1999, the term was included in the Oxford English Dictionary, which defined it as "the dissemination of false information by an organization in order to create a public image of environmental responsibility; also, the public image of environmental responsibility attributed to an organization, but considered to be without basis or intentionally misleading" (Seele and Schultz 2022).

The GWB has recently increased and gradually become a worrisome behavior, attracting scholarly attention from a wide array of disciplines—including business economics, environmental science, and social science—as well as from governmental and non-governmental organizations (NGOs). This interdisciplinary interest has led to a proliferation of

definitions and conceptual perspectives. As a result, there remained no universally accepted definition of GWB, and the term continued to evolve alongside growing societal concern and regulatory engagement (Martinez et al. 2020).

To capture the diversity of definitions, we reviewed the selected literature and identified 16 definitions from both academic and non-academic sources, as shown in Table 4. It can be observed that the understanding of this environmental opportunistic behavior has expanded today. This expansion in understanding is reflected in the evolution of GWB definitions, from the previous focus on "organizations creating a favorable environmental image through targeted environmental statements" to "organizations intentionally manipulating business practices in the sustainability disclosure process to gain organizational legitimacy in the context of social responsibility". Furthermore, from a stakeholder perspective, GWB definitions have also evolved from the corporate level to the government level.

Through comparative analysis of these 16 definitions, we identified four core conceptual elements of GWB, which are: (1) evade the crucial point; (2) selective disclosure; (3) decoupling; (4) deceptive manipulation. These elements align with previous attempts to categorize GWB in the literature. However, as shown in Table 4, GWB still lacks a universally consistent definition, consistent with previous discussions by scholars (Nemes et al. 2022). A key reason lies in the absence of clearly defined responsible entities, and different stakeholders may have different ways of describing and understanding it (Testa et al. 2018).

Through the review of the aforementioned collection of papers, regardless of the field, most research on GWB involves various relevant groups such as government, corporations, consumers, NGOs, advertising agencies, and media. Additionally, according to stakeholder theory, we include the natural environment in our research because it can be considered as an affected party by GWB (Wheeler and Maria 1998). Therefore, the definitions of GWB by different participants are shown in Table 5.

In summary, regardless of the degree of falsehood in corporate green communication, GWB exists only when other stakeholders emphasize the information. Therefore, it is feasible to define GWB based on the perspective of stakeholders as it is seen as a phenomenon from an outsider's point of view.

Table 4 Core conceptual elements of GWB

Classification	Source
1. Evade the crucial point	Guo et al. (2017), Merriam-Webster (2018), Arouri et al. (2021), Megura and Gunderson (2022), Li et al. (2022)
2. Selective Disclosure	Delmas and Burbano(2011), Mahoney et al. (2013), Marquis et al. (2016), Mateo et al. (2022)
3. Decoupling	Roulet and Touboul (2015), Siano et al. (2017), Yang et al. (2020),
4. Deceptive manipulation	Laufer (2003), Tregidga (2017), Kassinis and Panayiotou(2018), Farooq and Wicaksono (2021), Seele and Schultz (2022)

Table 5 Definition of GW from the stakeholder perspective

Stakeholder	Definition
Natural environment	Whether the behavior decisions of governments and corporation have negative effects on the environment
Government	1. Whether the behavior decisions of the enterprise violate environmental laws and regulations, and whether there are fraudulent behaviors 2.Whether the actual behavior of the government consistent with environmental policies, and whether there are fraudulent behaviors
Corporation	1. Whether the green marketing methods of the enterprise are authentic, and whether there is a gap between the substantive and symbolic nature of the behavior 2. Whether the green words, deeds, and standards of multinational enterprises unified
Consumer	Is there any misleading or deceptive behavior in the green promotion of enterprises
Advertising agency	Does marketing creativity objectively and truthfully describe the true environmental protection situation of enterprise products or services
Medium	Whether to report the true environmental protection information of the enterprise and whether to effectively supervise the environmental issues of the enterprise
NGOs	Whether to become supervisors and exposers of GWB for governments and businesses
Network platform	Whether environmental actions such as carbon trading and carbon auditing have positive effects on the environment, and whether there is any fraudulent behavior involved

Driving factors and underpinning theories of GWB

Through the review of literature, it was found that there are many factors influencing GWB, such as the characteristics of actors, complex psychological traits, external environment, etc. Based on the driving factor framework proposed by Delmas and Burbano (2011), the articles on driving factors are classified, as shown in Table 6. The research on mezzo-level driving factors of GWB is the most prominent (36.51%). Among all the driving factors studied, information communication of companies/organizations, non-market external environmental factors, and market external environmental factors ranked in the top three, accounting for 19.05%, 14.29%, and 11.11%, respectively.

The phenomenon is closely linked to certain theories. Stakeholder theory (22.27%), legitimacy theory (18.07%), institutional theory (16.39%), and signal theory (13.03%) are the most commonly used theories to explain GWB, as shown in Table 7. The following will elaborate on the relationship between these theories and the causes of GWB, and their importance. The theories presented should not be viewed as an exhaustive list, but rather as a starting point to broaden the scope of GWB research.

The following discusses the top three theories as analyzed from the literature.

First, stakeholder theory has been utilized by scholars to explain the GWB, which is influenced by non-market environmental factors such as government environmental policies and regulatory effectiveness (Delmas and Burbano 2011). For example, in highly polluting industries, a relaxed regulatory environment is one of the important reasons for triggering GWB (Li et al. 2024). When NGOs and social media are less ambitious about corporate environmental performance, it can contribute to GWB (Partzsch et al. 2019). These insights emphasize how changes in non-market environmental factors directly impact the pressure on stakeholders, thereby affecting the form of GWB in various industries. Moreover, stakeholder theory also suggests that in addition to non-market environmental factors, understanding why businesses engage in GWB is crucial by examining factors related to consumers, investors, competitors, and others (Feng et al. 2022). Stakeholder theory indicates that

Table 6 Distribution of articles on driving factors

Level	Drivers	No	Total	Percentage (%)
External environment (Macro)	1. Non-market external environmental factors (such as policy environment, NGO or media)	34		12.93
	2. External market environmental factors (stakeholders)	30		12.71
	Total		64	
Organizational environment (Mezzo)	1. Organizational characteristics	26		9.89
	2. Information Communication for organizations	42		15.97
	3.Organizational culture, incentive structure, etc	23		8.75
	4. Organizational inertia	27		10.27
	Total		118	
Individual environment (Micro)	1. Optimistic bias	15		5.70
	2. Myopic motivation	14		5.32
	3. Environmental responsibility awareness	11		4.18
	Total		40	
Mixed	Not just analyzing from one aspect	41	41	15.59
Total		263	263	100

Table 7 Theoretical research distribution of driving factors

Theory	No	Percentage (%)	Theory	No	Percentage (%)
Institutional Theory	39	16.39	Signal Theory	31	13.03
Social and Political Theory	12	5.04	Social Responsibility Theory	15	6.03
Stakeholder Theory	53	22.27	Attribution Theory	18	7.56
Resource Dependency Theory	13	5.46	Agent Theory	9	3.78
Legitimacy Theory	43	18.07	Attitude-Behavior-Context theory	5	2.10
Fraud triangle theory	4	3.23	Transaction cost theory	2	1.61
Total: 242					

stakeholders, such as investors, participate in corporate decision-making with dual objectives to demand CSR fulfillment and to gain returns through capital operations (Farooq and Wicaksono 2021). Considering these two objectives, companies face pressure from both investors and consumers during their development process, and the greater the pressure for environmental requirements, the more likely companies are to engage in GWB, especially in situations where there are minimal regulatory consequences. Besides, companies may engage in GWB to gain environmental legitimacy and meet stakeholder demands to avoid lagging behind competitors (Marrucci et al 2022).

Secondly, legitimacy theory suggests that organizations strive to gain social recognition for their actions in specific environments in order to ensure the continuity of their operations (Matejek and Gossling 2014). Therefore, companies adopt various means to legitimize their business activities under different environmental pressures in order to maintain their so-called "legitimacy of operations" (Minh et al. 2022). In this process, companies can seek regulatory legitimacy through GWB, gain cognitive legitimacy by actively promoting their environmental performance symbolically to earn societal trust (Partzsch et al. 2019); and attain normative legitimacy by imitating GWB strategies of other companies (Seele and Schultz 2022). In other words, the legitimacy theory provides a form for companies to seek legitimacy in relation to GWB in different environments, such as regulatory legitimacy, normative legitimacy, and cognitive legitimacy. Besides, selective disclosure, as one of the forms through which businesses seek legitimacy, is considered by legitimacy theory and stakeholder theory to be an indirect factor. However, signal theory views it as a direct factor.

Last but not least, companies are open systems that require continuous information and resource exchanges with their surrounding environment, allowing for mutual permeation. The institutional environment consistently constrains the strategic choices of the company (Marquis et al. 2016). Therefore, the institutional environment is a focal point emphasized by institutional theory. Due to different institutional pressures faced by organizations, such as regulatory environments, industry norms, and social expectations, companies also respond differently to them. Not all companies adopt substantive actions to navigate institutional pressures (Zhang and Sun 2021). When dealing with changes in the institutional environment, companies may choose to manipulate institutions, employing symbolic means to make surface-level changes in order to gain trust (Zhang 2022). In order to effectively control and adapt to the environment, top managers often provide intentional or unintentional explanations for social behavior in the surrounding environment (Gosselt et al. 2019). This is exactly how GW resulting from managerial traits can be explained within institutional theory.

Governance strategies of GWB

GWB governance research primarily focuses on corporate self-improvement (33.51%), government regulation (14.36%), and social supervision (12.77%), among other aspects. The research methods employed are diverse and relatively mature, as shown in Table 8.

Government regulation has been proven to be the most direct means of managing opportunistic behavior, and its regulatory mechanisms are considered to play a critical role in preventing corporate fraud and maintaining stable market development (Sun and Zhang 2019). First, government regulation can effectively reduce organizational misbehavior (Zhu et al. 2023), increase the likelihood of organizations taking substantive actions (Mateo et al. 2022), and influence the internal governance process of organizations (Wu et al. 2020). Second, government regulation can effectively mitigate the adverse effects of corporate social responsibility communication mechanisms, such as third-party green certification and sustainability ratings (Parguel et al. 2011). Additionally, governments use reliable assessment procedures to control green certification bodies and reduce the likelihood of a "lemon market" (Flagstad et al. 2022). We can conclusively state that green certification also plays a positive role in limiting GWB. However, according to Garrido et al. (2020), mandatory green certification regulations

Table 8 Distribution of GW governance literature

Category	General content	No	Percentage (%)
Government	Policy regulation, etc	27	14.36
Corporation	Social responsibility, brand influence, information disclosure, corporate environmental strategy, and project manager leadership, etc	63	33.51
Third-party certification	Green certification, ecological labeling, etc	19	10.10
Society	External stakeholder supervision, such as NGOs, media, consumers, etc	24	12.77
Technical method	Block-chain, Internet of Things, indicator measurement, etc	12	6.38
Mixed	More than one aspect	43	22.87
Total		188	100

may hinder the transmission of corporate information, leading to negative consequences. This suggests a potential trade-off, while regulations standardize practices, they may also reduce firms' incentives for beyond-compliance environmental innovation. Lastly, enhancing government regulatory capabilities can prevent GWB caused by collusion and conspiracy among stakeholders (Liu et al. 2023). However, excessive regulatory pressure may increase the likelihood of collusion among entities rather than eliminate it, especially in industries with high monitoring costs (He et al. 2020). Hence, while regulation is vital, its design and enforcement require careful calibration.

Corporate governance is the most crucial means to curb GWB, and corporate-level GWB governance is more important than national-level governance (Yu et al. 2020). With the increasing prevalence of CSR, this voluntary behavior driven by obligation promotes the spread of GWB (Gatti et al. 2019). However, the voluntary nature of CSR creates inherent limitations, firms may prioritize impression management over substantive action, engaging in "decoupling" between sustainability discourse and operational practices (Lubloy et al. 2025). Therefore, scholars have conducted governance research on GWB based on CSR from various perspectives, including different industries (Flagstad et al. 2022), different types of enterprises (Wu et al. 2020), and different stages of corporate development (Neumann 2021). Furthermore, governance research on GWB based on CSR includes, but is not limited to, aspects such as corporate development strategies, brand management, disclosure reporting, information quality, supply chain management, and managerial leadership. These multidimensional approaches reflect CSR's complexity but also reveal coordination challenges. These aspects of corporate operations constantly reflect CSR and also influence the occurrence of GWB. In our opinion, the best strategy for a company is to "do as you say". This type of internal governance requires robust internal monitoring systems to prevent the gap between words and actions that caused by competing organizational priorities.

Social supervision as an important external oversight force to ensure CSR, has attracted widespread attention in environmental governance. Firstly, Miller (2006) proposed that non-governmental organizations and the media should play the role of market watchdogs by investigating and publishing lists of GWB companies. For example, the "Chinese Greenwash Ranking" released by Southern Weekend has had a deterrent effect on GWB enterprises (Du 2015). Secondly, the thriving development of social media has strengthened society's scrutiny of GW. It is currently considered a powerful and crucial platform for corporate disclosure. Misleading or deceptive disclosures by companies can quickly be exposed and spread rapidly on the internet by other users, ultimately backfiring on the company (Amin et al. 2022). However, social media is a double-edged sword, as companies may deliberately reduce the disclosure of environmental information due to concerns about potential backlash. Lastly, governance strategies of social supervision also play a positive role. For example, social supervision strategies based on naming and shaming (Taebi and Safari 2017). Popescu et al. (2021) expressed concerns about GW and suggested that the governance of GW should incorporate the perspectives of the general public to form reliable sustainability assessment methods. However, public participation models face challenges in reconciling lay perceptions with technical environmental metrics, potentially privileging visible over substantive issues (Agne and Renata 2025). In a word, social supervision also cannot be ignored in GWB governance.

In summary, the governance of GWB requires a coordinated and multi-faceted approach. It should combine the compulsory nature of government regulations, the self-consciousness of CSR disclosure and the auxiliary role of social supervision to reduce the GWB. For example, existing research has shown that in the energy industry, a strict regulatory environment combined with active social supervision can significantly reduce GWB (Qu et al. 2025).

Classifications and consequence of GWB

Theoretical-oriented classifications

GWB takes different forms and has drawn interest from researchers in various fields. While scholarly research helps identify and categorize diverse manifestations of GWB, the list provided may not be exhaustive due to the ongoing exploration in this area. Laufer (2003) pointed out that GW encompasses three forms of behavior: concealment, confusion, and pretense. Since then, the manifestations of GWB have gained attention from scholars. According to the definition of GWB, one of the main reasons for its occurrence is the phenomenon of information asymmetry in green communication within organizations. Therefore, Amin et al. (2022) have distinguished GWB into four categories based on the degree of distortion in green communication: "false greenwashing," "potential greenwashing", "greenwashing", and "non-greenwashing". Aurélien (2017) has categorized GWB into three types based on the severity of its behavior: "light greenwashing," "hard greenwashing," and "deep greenwashing."

GWB can occur at the company level in terms of environmental practices for misleading consumers regarding the environmental benefits of products or services (Delams and Burbano 2011). Terms like "claim-based greenwashing" and "action-based greenwashing" have been mentioned (De Jong et al. 2018; Netto et al. 2020). Vollero et al. (2016) based on institutional theory classified GWB into two types: proactive and reactive. "Proactive greenwashing" can include hollow

green commitments, manipulating organizational practices to support one-sided or false claims about sustainability, and fabricating green labels (Vollero et al. 2016). "Reactive greenwashing" can include selective or false disclosures and exaggeration or false advertising (Matejek and Goessling 2014).

The aforementioned scholars have conducted fairly scientific classifications of GWB according to single classification methods. As research progresses, many scholars have also conducted studies on the combination of classification methods. Gatti et al. (2021) organically combined the motives for GWB and the levels of GWB, resulting in four types: fabrication, information selection, deceptive manipulation, and attention diversion. Torelli et al. (2020) combining factors from the field of GWB and levels of misleading environmental communication, have divided it into "corporate-level greenwashing," "strategic-level greenwashing," "product-level greenwashing," and "dark-side greenwashing." Pizzetti et al. (2021), taking into account the differences between "environmental talk and walk" in the supply chain, have distinguished GWB behavior into "direct greenwashing," "indirect greenwashing," and "substitute greenwashing." Therefore, GWB can occur in multiple forms.

Practical-oriented classifications

Different types of businesses can adopt various strategies and methods of GWB. Andre (2015) argues that welfare corporations serve as an antidote to GWB and are more prone to engaging in it compared to traditional corporations. On one hand, lax accountability standards foster GWB in welfare corporations, curbing their efforts towards social responsibility. On the other hand, welfare corporations lack independent audits. Kim et al. (2017) suggest that non-family firms tend to exhibit less proactive environmental actions contrary to their positive environmental slogans, and have a higher tendency towards GW compared to family firms.

From external corporate activities, GWB may be a wolf in sheep's clothing. Polluting corporations tend to engage in GWB through channels of corporate philanthropy (Du et al. 2016). Even conglomerates with better environmental governance mechanisms, their donation activities may not truly be charitable, serving as another form of GWB that hinders sustainable development (Choi and Hong 2022). In the context of carbon neutrality, Huckestein (2020) suggests that both companies and governments may use measures such as reforestation to offset greenhouse gas emissions in order to improve their image, rather than making a genuine contribution to climate protection. This inefficient public voluntary program is also one of the ways to GWB.

From internal corporate strategies, "confusing signals" can create misleading pro-environmental images. For example, fuel companies often use green imagery and rhetoric to mask environmental harm for profit maintenance (Megura and Gunderson 2022). Trading carbon allowances in carbon markets without strict carbon auditing processes can lead to GWB (Cao et al. 2022). Marrucci et al. (2022) argue that fashion companies redirect consumer attention from the environmental impact of leather manufacturing in their supply chain to communication strategies associated with the circular economy, a variant of circular washing as a GWB tactic. Tregidga (2017) suggests that environmental shadow reporting prepared by companies is largely a GWB process. Some companies may use third-party certification labels or standards to gain commercial legitimacy and enhance external reputation, misleading stakeholders to overlook negative aspects (Heras et al. 2020).

From other perspectives, GWB has evolved from exaggeration and inconsistency to forms that associate environmental concepts with health, employment, and other areas (Scanlan 2017). GWB has transitioned from "pseudo-green" practices to "pseudo-CSR actions" that collaborate with various illicit or unethical behaviors (Yu et al. 2020; Flagstad et al. 2022). Sterbenk et al. (2022) focusing on gender, argued that female empowerment advertisements often do not align with their broader corporate social responsibility activities, indicating the presence of GWB.

Regardless of the specific forms mentioned above, GWB can bring additional benefits to organizations or their executives, such as government subsidies (Huang et al. 2020), green premiums for products obtained through pseudo green (Kleffel and Muck 2023), and promotion to positions (Li et al. 2023b, a).

The consequences of GWB

From a market environment perspective, perceived GWB directly or indirectly has negative impacts on stakeholders. Firstly, GWB led to information overload, making it challenging for consumers to evaluate products and inhibiting them from making informed purchasing decisions (Kahraman and Kazancoglu 2019), thereby affecting their purchase intentions (Santos et al. 2023). Interestingly, Urbański and Ul Haque (2020) empirically found that consumer purchase intentions are not influenced by GWB. Regarding this contradictory result, we believe that GWB activates consumers' affective information processing mechanisms, where they still make purchases even when recognizing GWB, as affective persuasion fits their cognitive processing better than rational persuasion. Additionally, discrepancies between organizational green communication and actions could have negative effects on consumers' perceptions, such as green trust, perceived risk, and skeptical attitudes (Farooq and Wicaksono 2021). GWB from a brand could even influence consumers' willingness to purchase green products from other companies within the industry (Wang et al. 2020a, b).

Secondly, as a tactic, GWB could mislead investors' perception of a company's true performance (Pizzetti et al. 2021). Although different types of GWB have varied impacts on investors, they are equally destructive, eroding investors' confidence in environmentally friendly companies (Yang 2022). Furthermore, GWB could also lead to the premium effect (Wang et al. 2020a, b) and generate a "ripple effect" of imitation from competitors, imposing negative impacts on genuinely green companies (Testa et al. 2022).

Furthermore, from the perspective of the organization itself, GWB was negatively correlated with corporate financial performance and environmental performance (Du 2015), as it weakened brand reputation, trust, loyalty, market value and increases consumer confusion about CSR (Rahman and Bang 2022). Additionally, the risk impact of GW varies for different types of companies (D'Arcy et al. 2020). However, some scholars have indicated that there is no relationship between GWB and financial performance for banking firms (Wu and Shen 2013). These performance-related studies implied that organizations do not benefit from GWB. Pimonenko et al. (2020) calculated that a 1% increase in GW leads to a 0.56% decrease in a company's green brand using the PLS-PM modeling method. Furthermore, GWB could affect a company's moral culture, indicating that not all that glitters are gold (Szabo and Webster 2021).

Lastly, the negative impacts of GWB could also permeate within the organization's employees. Compared to external stakeholders, employees might be the first to identify discrepancies between corporate CSR communication and actions and reacted to GWB (Schons and Steinmeier 2016). Recent research has found that such discrepancies indeed led to negative effects on employees, including emotional exhaustion, job turnover, and reduced personal engagement in voluntary social responsibility activities (Scheidler et al. 2019; Li et al. 2022). These circumstances resulted in employees losing trust in their organization, triggering a cynical attitude among employees, ultimately leading to decreased job performance.

Discussions

5.1 Research trends and GWB in different industries.

The overall number of articles related to GWB is gradually increasing, and the fields involved are also becoming more diverse. Most of them are concentrated in the manufacturing, service, and energy industries. However, the knowledge related to GWB varies among different industries. Based on the basic information in Table 2, there may be several reasons leading to the differences in the three sectors.

Firstly, GWB does not have a relatively clear definition. The concept of GWB cannot be easily described in every industry. For example, many existing studies conceptualize GWB from a marketing and communication perspective (Yang et al. 2020; Nemes et al. 2022), which may be suitable for consumer-facing sectors but inadequate for upstream industries like manufacturing, where GWB may take the form of technical misreporting, data manipulation, or perfunctory environmental rectification. This ambiguity in definition poses challenges to research and practice. From a research perspective, it has led to fundamental differences in the explanations of GWB driving factors among different schools of thought. The institutional school emphasizes the lack of regulation, while the behavioral school focuses on cognitive biases. From industry practice, the lack of clear standard definitions may provide opportunities for enterprise GWB. For example, in the energy sector, companies may exaggerate emission offsets without standardized validation protocols, while in the service industry, vague sustainability statements may lack measurable standards. There seems to be a consensus that the GWB concept endows industry characteristics into practical cognition with appropriate scenarios. Although there are many literatures on GWB in specific industries, further discussion is still needed on the concepts and standards of GWB in specific industries.

Secondly, there are different industry backgrounds. The financial industry provides intermediary services for the flow of funds for enterprises and offers various financial products. As shown in Table 2, one way for GWB in the financial industry is that the funds raised through green financial products are not used for green projects. This can indicate that the financial industry can have a dependency relationship with other industries in GWB. Especially in high polluting industries, because green finance promotes the greenwashing of high polluting enterprises. Another manifestation is nominal green financial products. Danilov (2022) believes that the increase in sustainable financial instruments can also lead to an increase in GWB, which may be because green finance is in its early stages and its definition, standards, and regulation are not yet mature, providing issuers with an opportunity to conduct GWB through packaging (Zhu et al. 2023). The manufacturing and service industries are essentially customer-oriented in providing corresponding products or services. Therefore, in the face of various pressures, GWB is mainly carried out through misleading marketing methods. But the difference between manufacturing and service industries is the process of GWB occurring, while the service industry only appears after consumer consumption (Chen et al. 2019). However, there is a lack of research on more detailed manifestations of GWB in the service industry based on the timing of its occurrence. Therefore, the above two points can serve as explanations for the differences in different industries.

In addition, this article also emphasizes the construction industry, as it is the fastest growing industry since 2016. Compared to other industries, the construction industry has

attracted the attention of many scholars to the GWB in this field due to its high complexity, high uncertainty, and dense contractual relationships. There are numerous stakeholders and their interaction processes endow the industry with different situational patterns, such as decision-making interactions between owners and contractors, contractors and supervisors, government and owners, and other stakeholders. This creates more imaginative space for the occurrence of GWB, such as collusion. The GWB generated by opportunistic behavior-chain coupling is also a manifestation that has not been mentioned in other industry literature (He et al. 2020).

The manifestation forms of GWB

By referring to the literature findings, seven core elements of seven GWB's manifestation forms have been found according to their performance characteristics. The GWB's manifestation forms evolve around four main aspects: economy, society, environment, and politics as illustrated in Fig. 5. The details of seven core elements are as follow.

(1) Do things carelessly. It means that organizations handle environmental work carelessly, simply aiming to get by. This attitude of indifference in the organization's work can lead to the occurrence of GWB and, over time, form a negative work memory, making GWB a habit. For example, in engineering construction, contractors respond perfunctorily to environmental rectification issues.

(2) Deliberate concealment. It means that organizations selectively hide negative environmental information to create a pro-environmental corporate image and meet their own development goals. For example, Coca-Cola company claims to reduce water consumption by 4% annually to conserve water resources, but it remains silent on its actual water usage.

(3) Divert attention. It means organizations aim to divert stakeholders' attention to cover up their unethical behavior. This can be done through third-party certifications in the absence of detailed information and credible authentication, or through the use of vague and hollow language to explain products or behaviors. It can also involve using inefficient environmental activities to improve their social image rather than focusing on environmental protection. For example, Huckenstein (2020) believes that companies use methods such as afforestation to compensate for greenhouse gas emissions in order to improve their image, rather than contributing to climate protection.

(4) Double standards. It means that organizations adopt inconsistent behaviors with the original standards for the same nature of affairs, especially when there is a lack of strict accountability standards. This is often done by lowering work standards to meet the production needs of managers. For example, multinational corporations may adopt different standards for a business in different regulatory environments.

(5) Practice fraud. It means that sustainable communication of organizations creates intentional manipulations in business practices, satisfying the organization's perception of being environmentally friendly through false disclosure of information. This allows the organization to gain higher economic benefits and social reputation through fraudulent actions. For example, Huang et al. (2020) mentioned that emissions cheating is a common practice among many

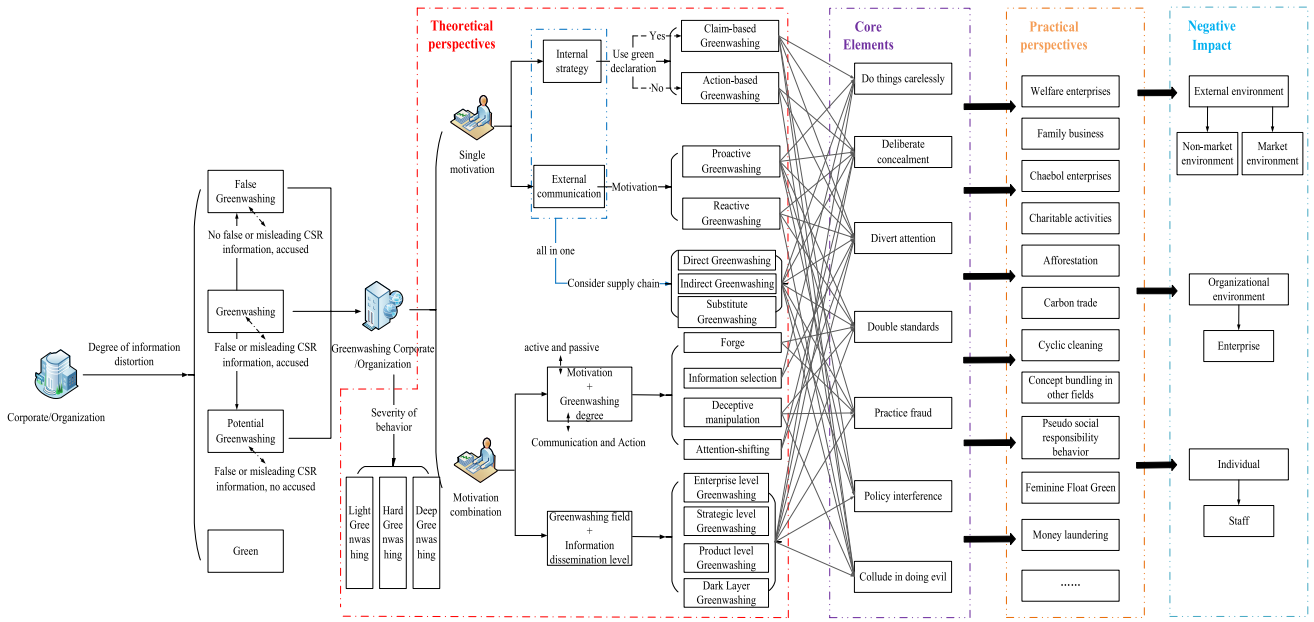


Fig. 5 The sub-framework of GW based on GWB's manifestation forms

giants in the automotive industry during Volkswagen's GWB scandal.

(6) Policy interference. It means that organizations manipulate and disrupt institutions to gain legitimacy when facing institutional pressures. This is not only done passively in activities but also aligns with the perspective of institutional theory. Both companies and governments may engage in such behavior under performance pressures. For example, real estate developers create regulatory blind spots and lower construction standards by leveraging their relationships with local policy makers.

(7) Collude in doing evil. It means that GWB will be accompanied by opportunistic behavior from other stakeholders in its dynamic evolution, characterized by strong concealment. GWB differs from the six core factors mentioned above where it emphasizes the scenario of interaction. organizations may take the initiative to lure other entities into opportunistic behavior (such as collusion, regulating prisoners of war), creating more convenient conditions for GWB to gain more benefits. For example, the contractor of the Zhengwan high-speed railway project has repeatedly violated construction regulations, and their confidence in sustained greenwashing comes from collusion with the owner and supervisor (He et al. 2020).

The theoretical framework of GWB

By reviewing the four dimensions of GWB mentioned above, it can be condensed into three aspects (i.e. people, purpose, and process), which can form a 3P framework to facilitate a better understanding of GWB, as shown in Fig. 6. The 3P framework has been applied in the field of environmental management to achieve sustainable development of organizations (Wang et al. 2020a, b). As an opportunistic behavior in environmental management, in-depth analysis of GWB from different dimensions is beneficial for improving the organization's environmental management level, which is in line with the 3P framework. The term "people" refers to the actors involved in GWB, and GWB is a phenomenon observed by bystanders. The term "process" refers to why organizations engage in GWB, the methods they employ, and the potential consequences. The term "purpose" refers to the governance strategies of GWB aimed at transforming it into true green.

The definition of concepts can abstract and summarize the essential characteristics of things, forming conceptual thinking inertia. The core elements of the GWB definition summarized in Table 3 can be regarded as the connotation of the GWB's concept, which reflects the unique

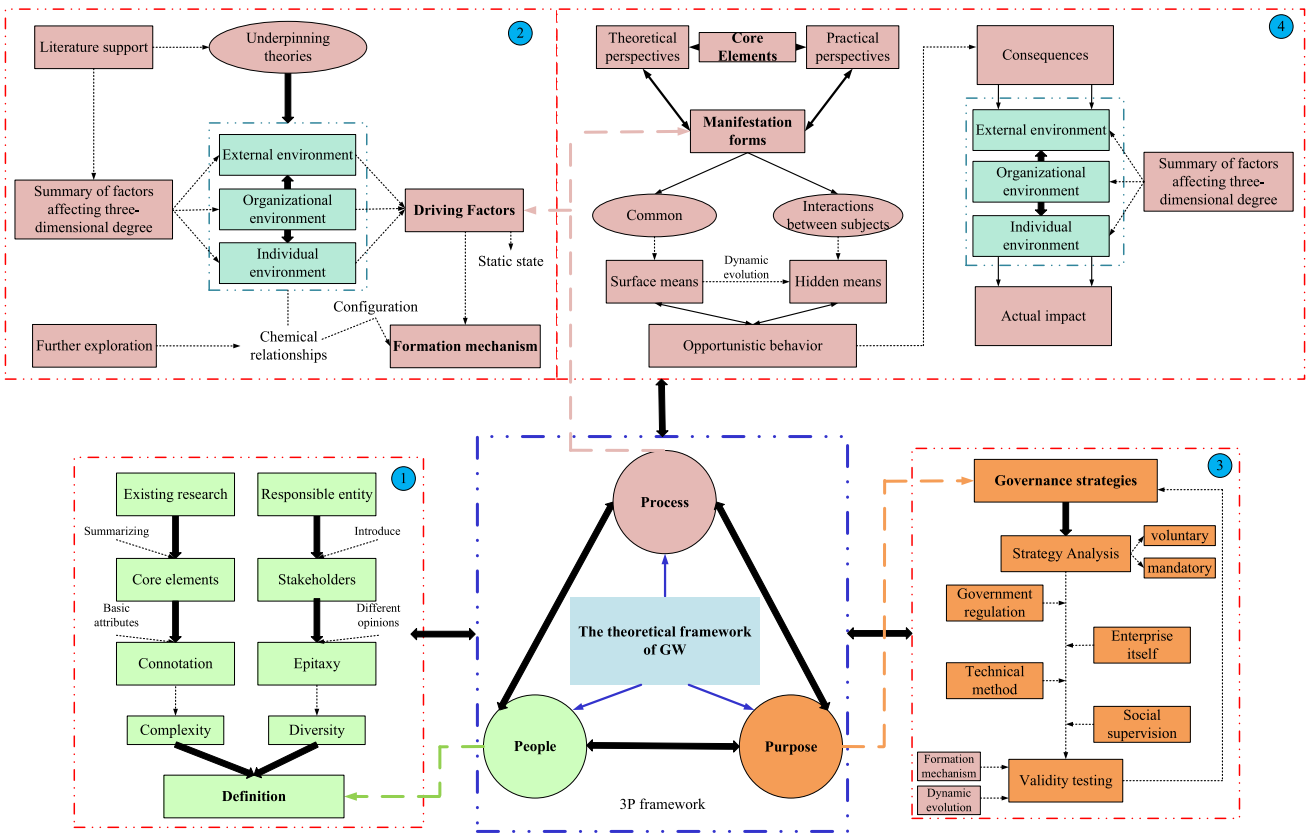


Fig.6 The theoretical framework of GW

attributes of GWB and also demonstrates the complexity of the GWB's concept, but it can still be seen that it does not have a strict definition. As a type of organizational development strategy, GWB is inevitably associated with stakeholders. Table 5 defines the concept of GWB from a stakeholder perspective, emphasizing both the diversity of its concepts and the extensiveness of the concepts, namely the scope of the objects reflected by the GWB's concept. As academic research on this topic deepens, its concept will gradually expand.

Various supporting theories help to clarify the driving factors of GWB, especially from the macro external environment to the micro individual level, but theoretical exploration cannot be separated from the theoretical development of economics and management. Based on the perspective of neoclassical economics, in order to pursue profit maximization, the GWB can help organizations deceive government subsidies and win market favor. From the perspective of information economics, information asymmetry can provide operational space for an organization's GWB. They can exploit these gaps by selectively disclosing or exaggerating environmental achievements, thereby gaining unwarranted legitimacy or financial advantages. From the perspective of stakeholders in organizational management, the pressure and attitudes of internal and external stakeholders within an organization are also important factors that trigger GWB. For instance, when influential stakeholders such as institutional investors place strong emphasis on environmental performance, firms may feel compelled to appear compliant. For regulatory agencies or managers seeking action to reduce the occurrence of GWB, understanding the driving factors of GW is crucial in order to determine appropriate strategies to resist it.

The governance of GWB is the key to whether enterprises can achieve green and sustainable development. The governance strategies proposed from different perspectives have their respective advantages in curbing prominent issues in corporate environmental management in Table 7. It is worth noting that existing research often conducts localized analysis of the governance strategies adopted by a certain type of entity, which may lead to local optima and overall suboptimal situations. In addition, the effectiveness testing of strategies should also be combined with the characteristics and laws of GW, especially the formation mechanism and dynamic evolution of GWB.

From the seven core elements in Fig. 5, it can also be seen that GWB has gone from perfunctory management work to blatant deception, to increasingly covert management methods, and finally constantly touches on the dynamic evolution process of legal and regulatory ambiguity and vacuum zones. Regardless of the form of expression, GWB is essentially an opportunistic behavior.

Further explanation of different research dimensions

Concept

Concepts are crucial for understanding the GWB. Table 4 clearly shows that existing research has expanded the understanding of GWB from misleading statements to deceptive manipulation, and revealed the four core elements of GWB definition. However, due to the diversity of disciplines and the lack of clear definition of responsible parties, it reflects the lack of a strict definition. Therefore, this study defines GWB from a stakeholder perspective, because it is a behavior in the eyes of onlookers. It is worth noting that the government can serve as both the decision-maker of GWB and its affected party.

GWB as an organizational survival strategy, exists in various environments within society (Minh et al. 2022). Therefore, based on institutional theory and from the perspective of stakeholders, this article provides a relatively broad definition of GWB, which means that stakeholders, in response to certain work pressures, use information gaps to actively or passively engage in non-standard business behavior, regardless of whether it has a negative impact on the environment, in a morally detached manner. It can be seen that in the current context of sustainable, GWB has become one of the unsettling factors in ecological construction. The sustainable development of the environment should not only focus on environmental pollution behavior, but also on the GWB.

Driving factors

It is undeniable that research on the driving factors of GWB from the three dimensions of external environment, organizational environment, and personal traits has become relatively mature, as shown in Table 6. Scholars (Partzsch et al. 2019; Feng et al. 2022; Zhao et al. 2022) have filled many studies on influencing factors based on the driving factor framework proposed by Delmas and Burbano (2011). This framework has been widely cited and validated in subsequent research (He et al. 2020; Zhu et al. 2024), offering a structured lens through which to identify the antecedents of GWB. However, although the framework proposed by Delmas and Burbano still have influence, it is not without limitations. Some scholars have pointed out that although the categories of this framework were comprehensive, it might oversimplify the complex interactions between internal motivation and external pressure (Wang et al. 2025). Therefore, recent studies have attempted to improve this framework by incorporating industry-specific institutional logic or management cognitive frameworks (Wang et al. 2023a, b). These improvements indicate that although the original

framework provides a solid foundation, it must be flexibly applied to adapt to changes in context.

In addition, these factors present a static effect within its framework. Specifically, the generation of GWB is the result of the joint action of multiple influencing factors, but existing research usually analyzes the independent and net effects of each factor, while ignoring the chemical reactions generated by the interdependence between factors, that is, the formation mechanism. Compared to the study of influencing factors, the formation mechanism of GWB is complex and requires comprehensive integrated research. Although a few studies have begun to adopt the fraud triangle theory to analyze the formation mechanism of organizational GWB (He et al. 2022; Chen et al. 2022). But it is only at the micro level within the framework of driving factors. This also inspires scholars to further explore this topic in the future. In addition, there are a wide range of theories supporting the study of driving factors, as shown in Table 7. Therefore, it is not difficult to find a suitable theoretical framework for studying the formation mechanism of GWB, reasonably explaining the correlation between factor combinations and outcome variables, and revealing complex causal relationships such as asymmetry, concurrent causality, and equivalence.

Governance strategies

Table 8 shows that over 65% of GW governance research is mainly conducted from three aspects: internal improvement of enterprises, government regulation, and social supervision. However, while this growing body of research offers valuable insights, several limitations deserve attention. For instance, although government regulation is widely recognized as a deterrent to GW, its effectiveness is often constrained by institutional weaknesses, regulatory capture, or lack of enforcement mechanisms, especially in developing economies (Marquis et al. 2016). Similarly, the role of social supervision is frequently praised in theory, yet empirical evidence remains fragmented. While media exposure and NGO activism can place pressure on firms, their influence is highly context-dependent and may be insufficient in industries with low visibility or weak stakeholder engagement (Nepal et al. 2025). In addition, other governance strategies, especially in some hot research areas such as block-chain technology and indicator measurement, have not been fully explored, and these sporadic studies are only the beginning of a long exploration.

At present, governance strategies for GW focus on a top-down governance framework and institutional design, which they often rely on a single government strategy (He et al. 2020). If combined with the characteristics of GW in different industries, its applicability and operability still need to be explored. For example, temporary organizations in the construction industry, compared to permanent organizations such as manufacturing, have strong externalities in their environmental issues due to their involvement of numerous stakeholders, limited lifecycle, and short-term cooperative relationships (Wang et al. 2023a, b). Therefore, its governance strategy should be designed with targeted measures based on its characteristics. Meanwhile, enterprises, governments, and the public have different target preferences and interest demands, and they have conflicts between short-term and long-term benefits, economic and environmental benefits, direct and indirect benefits. How to leverage the balancing and corrective effects of multiple stakeholders and tap into the potential of joint governance based on the characteristics and laws of GW is a practical and urgent issue that needs to be addressed. In addition, from Table 5 and Fig. 5, it can be seen that GW involves numerous stakeholders and diverse forms of expression. Therefore, dynamic prevention based on different behaviors and corresponding scenarios, and changing the motivation of stakeholders for GW, is another promising direction for future research.

Manifestation forms

GWB, as an opportunistic behavior in environmental management, manifests in various ways and is accompanied by the evolution of behavior. The manifestation forms of do things carefully, delegate consideration, divert attention double standards, practice fraud, and policy intervention can be considered as static core factors that reflect the performance characteristics of an organization's unilateral GWB. And include in doing evil can be seen as a dynamic core factor. On the one hand, it reflects that opportunistic behavior can be transmitted, evolved, and coupled through a series of behaviors in various stages of GWB. On the other hand, it also reflects the GWB created by behavioral alienation between subjects. The above forms a chain of opportunistic behavior. By combining real-life cases, we can better understand such situations. Taking construction projects as an example, due to the attention of government departments, the contractors of the Jingxiong Railway project carried out barbaric construction under the banner of green environmental protection and key projects. Their sustained GWB confidence comes from collusion and collusion with participating units (People's Transportation Network 2018). From this, it can be seen that this manifestation of collusion is more worthy of our attention. However, there is currently a lack of quantitative characterization of this feature, and the dynamic evolution of the opportunistic chain is not yet clear. Future research can explore the hidden manifestations of GWB in depth.

5.5 Research contributions.

The research has developed a comprehensive theoretical framework of GWB with clarifications on manifestation forms of GWB. This study has two theoretical contributions to the limited development and literature of GWB.

Firstly, compared to existing GWB research that mostly focuses on specific aspects such as the GWB's driving factors (Delmas and Burbano 2011), GWB's definitions (Bernini and La Rosa 2023), and GWB's consequences (Santos et al. 2023), this study utilized a system thinking approach to integrate above-mentioned GWB-related aspects, forming a comprehensive scientific understanding of GWB research and filling the research limitations of the lack of systematic. Although some studies have analyzed the GWB's manifestations forms from the perspective of organizational motivations, the theoretical research in this area is relatively scattered and lacks a connection with practical research. Therefore, this study builds a bridge between theoretical and practical research by summarizing the core factors of GWB manifestations, deepening the research on GWB manifestations. Finally, for the first time, a scientific and complete theoretical framework for GW was constructed using the 3P framework. This framework advances existing literature in three key areas: (1) the people dimension highlights the dynamic actor structure in GWB, (2) the process dimension depicts the mechanisms and evolution of opportunistic behavior, and (3) the purpose dimension proposes governance oriented strategies to mitigate symbolic inappropriate behavior. Meanwhile, the use of the proposed theoretical framework also emphasizes the potential research topics of GWB. This theoretical framework can provide valuable assistance for managers in decision-making processes such as evaluation, management, and supervision, enabling them to effectively narrow the gap between substantive and symbolic behaviors. For example, it assists managers in identifying early warning signals of symbolic behavior through actor analysis. Develop targeted prevention and governance strategies based on the formation mechanism of GWB in specific industries.

Secondly, although some studies have defined some basic principles of GW, we have further integrated clear responsibilities of stakeholders based on the research of scholars and institutional theory, which is different from the previous definitions of GWB (Tregidga 2017; Yang et al. 2020; Li et al. 2022). This not only reflects the connotation of GW definition, but also its extensibility. Secondly, more specifically on GWB, by refining the core viewpoints of GW definitions and conducting a dual analysis from theoretical and practical perspectives, this study reveals new core elements of GWB manifestation forms from the static behavior to dynamic evolution, such as double standards, practice fraud, deliberate concealment, divert attention, policy intervention, do things carelessly and collude in doing evil, which are different from the elements of GWB manifestations summarized by Yang et al. (2020). As a result, by establishing a connection between theoretical research and practical behavior through the core elements of GWB's manifestation forms, this research can assess hidden practices of GWB more effectively and current limited literature of GWB.

Managerial implications

The management implications of this study are as follows. Firstly, the key factors of GWB should be identified based on the characteristics of different industries (such as permanent and temporary organizations), and the formation mechanism of GWB should be clarified, in order to develop targeted governance strategies. Secondly, it is necessary to strengthen the regulatory role of the government, formulate detailed and unified standards for defining greenwashing and environmental regulatory policies, and help stakeholders identify GWB. Meanwhile, it is imperative to strengthen social supervision and form a multi-party collaborative governance strategy. Finally, establish a flexible and differentiated certification system, such as ISO14000, green building, should be implemented. Through mutual cooperation between different industries, we can jointly reduce the occurrence of GWB.

Conclusions, limitations and prospects

Conclusions

This study provided a comprehensive definition of GWB from the perspective of stakeholders, and based on this, a basic definition of GWB was formed by using institutional theory. It analyzed the underpinning theories of GWB's driving factors and integrated them into three aspects: external environment, organizational environment, and individuals. It introduced governance measures for GW from the perspectives of government, companies, and society. Additionally, it established a connection between the theoretical and practical research on GWB's manifestation forms by summarizing seven core elements, namely, do things carelessly, deliberate concealment, divert attention, double standards, practice fraud, policy interference, and collude in doing evil. These elements reveal GWB's evolution from superficial to deceptive and covert tactics. As a result, a complete theoretical framework of GWB has been developed based on the "3P" perspective, which fills the gap of the current research lacking systematic induction of GWB. The proposed framework offers a new perspective for sustainable development in the environment field.

limitations and prospects

Certain limitations need to be considered in this research. Firstly, this research considered the WOS database, which is a reliable source of data. However, we acknowledge that some publications may still be missed, which could affect the theoretical sources included in our analysis. Therefore, it is possible to combine data from other databases to enrich the information for a better study of GWB. Therefore, data from other databases can be combined to enrich information for better research on GWB. Secondly, although our theoretical framework is universal, the formation mechanism of GWB may exhibit different characteristics depending on the industry (such as the construction industry with temporary features). Therefore, future research can use qualitative case analysis or industry based surveys to explore how these unique conditions shape GWB, in order to validate and refine our 3P framework in different practical contexts. Finally, this study only selected three industries for differential analysis of GWB related content. Future research can classify different industries and propose a comprehensive comparative analysis and insights on GWB among different industries.

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Declarations

Conflict of interest: The authors declare that they have no conflict of interest.

Human or animals participants This article does not contain any studies with human participants performed by any of the authors.

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